

A RAG-Based System for Pharmaceutical Decision Support in the Portuguese Regulatory Context

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Abstract—Pharmaceutical practice depends on rapid and reliable access to a dense and constantly evolving body of technical and regulatory documentation. Large Language Models (LLMs) offer a promising natural-language interface to this information, but suffer from two well-known limitations that make them unsuitable for direct clinical use: static post-training knowledge and hallucination. This paper presents the design and implementation of a Retrieval-Augmented Generation (RAG) system for pharmaceutical decision support in the Portuguese context. The system combines hybrid retrieval with natural language generation, grounding responses in official national documentation regulated by INFARMED, with full source traceability. The adopted architecture integrates Advanced RAG, Modular RAG, and Corrective RAG (CRAG) paradigms, implemented with Claude 3.5 Sonnet, Gemini Embedding 2, and the Qdrant vector database, orchestrated via LangChain and exposed through a FastAPI REST interface. The work addresses three gaps identified in the literature: operation entirely in Portuguese over INFARMED-regulated documentation, built-in compliance with the GDPR and the EU AI Act for high-risk systems, and retrieval evaluation and correction mechanisms inspired by CRAG. System performance is evaluated using the RAGAS framework, measuring faithfulness, answer relevancy, context precision, and context recall in scenarios representative of real pharmaceutical usage.

Index Terms—Retrieval-Augmented Generation, Large Language Models, Pharmaceutical Decision Support, INFARMED, EU AI Act

I. INTRODUCTION

The pharmaceutical sector is characterised by a density of technical and regulatory information with few parallels in other domains. Pharmacists and pharmacy technicians must quickly and reliably access information about drugs, interactions, dosages, contraindications, and therapeutic protocols scattered across a vast documentary corpus — package inserts, monographs, therapeutic guides, hospital formularies, and INFARMED regulations. This challenge is compounded by the constant evolution of pharmacological knowledge: new drug approvals, guideline revisions, and safety alerts are published regularly, making static information systems rapidly outdated.

LLMs have demonstrated the ability to understand clinical questions in natural language and generate contextually appropriate responses. However, their direct adoption in pharmaceutical contexts is hindered by three key limitations: (i) knowledge becomes static after training; (ii) responses lack source

attribution, making verification impossible; and (iii) models can generate plausible but factually incorrect statements — a phenomenon known as hallucination [1]. These limitations are particularly critical in high-stakes domains such as pharmacy, where an incorrect dosage or contraindication can directly harm patients.

The Retrieval-Augmented Generation paradigm addresses these limitations by grounding model responses in a dynamically queried external knowledge base [1], enabling source traceability and access to up-to-date documentation without retraining. Empirical evidence confirms that specialised RAG systems built over structured pharmaceutical documentation outperform both generic LLMs and non-specialised RAG implementations on clinical decision support tasks [2].

Despite this progress, the vast majority of existing biomedical RAG systems rely on English-language public knowledge bases, with little adaptation to multilingual contexts or national proprietary documentation [3]. This gap is particularly relevant for the Portuguese pharmaceutical context, where clinical documentation, INFARMED regulations, and national guidelines follow distinct linguistic and regulatory conventions. Moreover, systems deployed in the EU must comply with the EU AI Act, which classifies clinical decision support systems as high-risk AI, imposing requirements of transparency, traceability, and human oversight that are largely absent from systems described in the literature [4].

The present paper makes the following contributions:

- A RAG system operating entirely in Portuguese over INFARMED-regulated documentation, addressing the identified gap regarding multilingual biomedical RAG;
- An architecture combining Advanced RAG, Modular RAG, and CRAG, with hybrid retrieval and LLM-as-Judge reranking tailored to pharmaceutical documents;
- Built-in GDPR and EU AI Act compliance, including source traceability, mandatory disclaimers, and audit logging in every generated response;
- Evaluation using the RAGAS framework in scenarios representative of real Portuguese pharmaceutical usage.

The remainder of this paper is organised as follows. Section II provides the relevant theoretical background. Section III reviews related work. Section IV describes the system architecture and implementation. Section V presents the evaluation

methodology and results. Section VI concludes the paper and outlines future work.

II. BACKGROUND

A. Retrieval-Augmented Generation

Modern LLMs are built on the Transformer architecture [5], which uses self-attention to model long-range dependencies in text. RAG [1] extends LLMs with a retrieval mechanism: given a user query, the system first retrieves relevant passages from an external knowledge base and then conditions the LLM response on the retrieved context. This three-step pipeline — retrieve, augment, generate — allows the model to access up-to-date, domain-specific information without retraining and to attribute each response to specific source documents.

The evolution of RAG architectures has progressed from the initial *Naive RAG*, which performs a single-step vector similarity retrieval, to *Advanced RAG*, which introduces query reformulation, hybrid retrieval, and reranking [6]. *Modular RAG* further organises the pipeline into independent, replaceable components [6]. *Corrective RAG (CRAG)* adds a quality evaluation step over retrieved documents, triggering a corrective retrieval cycle when relevance falls below a defined threshold [6]. More recent developments include *Self-RAG* and *Agentic RAG* [7], which introduce autonomous retrieval decisions and multi-step reasoning respectively.

B. RAG Evaluation with RAGAS

The RAGAS framework [8] provides reference-free automatic evaluation of RAG pipelines across four complementary metrics. *Faithfulness* measures whether claims in the generated response are supported by the retrieved context — critical in clinical settings where unsupported statements can induce therapeutic errors. *Answer relevancy* evaluates whether the response pertinently addresses the original question. *Context precision* assesses whether retrieved chunks are relevant to the query, penalising the inclusion of low-utility fragments. *Context recall* measures whether the retrieved context covers all information necessary to correctly answer the question.

C. Regulatory Framework

The GDPR (Regulation EU 2016/679) classifies health data as a special category of personal data requiring reinforced protection, and mandates privacy-by-design principles, data minimisation, and the right to explanation of automated decisions.

The EU AI Act (Regulation EU 2024/1689) classifies clinical and pharmaceutical decision support systems as *high-risk AI*, imposing requirements of risk management (Art. 9), transparency and user information (Art. 13), human oversight mechanisms (Art. 14), and technical documentation and audit logging (Arts. 17 and 12). These obligations directly shape the system design presented in this paper.

III. RELATED WORK

A systematic literature review following PRISMA methodology [9] was conducted to identify existing RAG approaches for pharmaceutical and biomedical decision support. From an initial set of 51 articles, 14 were selected after duplicate removal and screening against predefined inclusion/exclusion criteria.

Evidence from the literature confirms the efficacy of specialised RAG over generic LLMs in biomedical tasks. A meta-analysis of 20 studies reported an odds ratio of 1.35 in favour of RAG-augmented systems compared to standalone LLMs [2]. Two studies are directly aligned with the present work.

Kim et al. [10] proposed QA-RAG, a framework with a dual retrieval mechanism applied to pharmaceutical regulatory compliance, combining semantic similarity search with keyword-based retrieval. This approach outperformed conventional RAG in precision, recall, and F1-score, motivating the hybrid retrieval design adopted in this system.

Borishetty et al. [11] developed PGxRAG, a RAG assistant built over structured pharmacogenomic guidelines (CPIC, FDA, KNMP), achieving 95.1% precision with GPT-4o. This validates the approach of grounding RAG systems in structured clinical knowledge bases.

The MIRAGE benchmark [12] evaluated 41 combinations of corpora, retrievers, and LLMs across 7663 medical questions, providing comparative baselines applicable to the evaluation of the present system.

A scoping review of RAG in clinical medicine [3] identified three recurring gaps: (i) dependence on English-language public datasets, (ii) embedding models centred on English, and (iii) insufficient evaluation of bias and safety. The present work directly addresses all three gaps in the Portuguese pharmaceutical context.

Bossenz et al. [13] evaluated eight embedding models and multiple chunking configurations on a hospital RAG system with 1219 real documents, demonstrating that chunking strategy significantly impacts retrieval quality — an empirical basis for the chunking parameters chosen in this system.

IV. SYSTEM ARCHITECTURE AND IMPLEMENTATION

A. Architectural Decision

Following analysis of available RAG paradigms against the system requirements and the pharmaceutical domain's specifics, a hybrid architecture combining **Advanced RAG**, **Modular RAG**, and **CRAG** was selected, as summarised in Table I.

Naive RAG was rejected due to its reliance on a single-step semantic retrieval, insufficient for pharmaceutical documents where exact drug name or active ingredient matching is frequent. Agentic RAG was rejected because its behavioural unpredictability is incompatible with the transparency and human oversight requirements of the EU AI Act for high-risk systems.

TABLE I
RAG ARCHITECTURE EVALUATION

Architecture	Decision	Reason
Naive RAG	Rejected	No hybrid retrieval or safety mechanisms
Advanced RAG	Adopted	Hybrid retrieval, reranking, validation
Modular RAG	Adopted	Modularity and maintainability
CRAG	Adopted	Retrieval evaluation and correction
GraphRAG	Rejected	Graph construction complexity out of scope
Agentic RAG	Rejected	Unpredictability, incompatible with EU AI Act
Self-RAG	Rejected	Requires supervised fine-tuning

B. System Overview

The system is organised into three functional layers, illustrated in Fig. 1. The *input and control layer* receives user requests through a REST API and applies safety guardrails to both inputs and outputs. The *orchestration layer* coordinates the RAG pipeline, managing vector retrieval, embedding generation, and response production. The *knowledge base layer* holds the indexed corpus of official Portuguese pharmaceutical documentation. An audit log runs transversally across all layers, recording every interaction for traceability and regulatory compliance.

C. Ingestion Pipeline

The ingestion pipeline processes and indexes pharmaceutical PDF documents into the knowledge base. Textual content is extracted using PyMuPDF, while tabular content (e.g., dosage schedules, pharmacokinetic parameters) is handled by pdfplumber, which preserves table structure before chunking. After text cleaning and normalisation, documents are segmented into chunks of 1000 tokens with a 200-token overlap between consecutive segments, following the empirical guidance of Bossenz et al. [13].

Each chunk is enriched with structured metadata — document type, filename, publication date, page number, and section — enabling metadata-filtered retrieval and precise source citation in generated responses. Chunks are then encoded by Gemini Embedding 2 and stored in Qdrant alongside their metadata. Visual content (chemical structures, pharmacokinetic plots) is currently discarded; multimodal extraction is identified as future work.

D. Query Pipeline

The query pipeline processes user questions at runtime through a sequence of quality-controlled stages, illustrated in Fig. 2.

Input guardrail. Incoming queries are first validated for pharmaceutical domain scope and inspected for prompt injection patterns. Out-of-scope or malicious queries are rejected with an informative message before accessing the retrieval pipeline.

Hybrid retrieval. The validated query is encoded by Gemini Embedding 2 and submitted simultaneously to a semantic similarity search and a keyword search in Qdrant. This dual-channel approach, motivated by the QA-RAG results of Kim

TABLE II
TECHNOLOGY STACK

Component	Technology	Rationale
LLM	Claude 3.5 Sonnet	Low hallucination rate; Constitutional AI
Embeddings	Gemini Embedding 2	Native PDF + multilingual (100+ languages)
Vector DB	Qdrant	Hybrid retrieval; metadata filtering; Docker
Orchestration	LangChain	Mature RAG support; modular LCEL
API	FastAPI	High-performance Python; auto-docs
Deployment	Docker	Reproducible local deployment (GDPR)

et al. [10], captures both the semantic intent of the query and exact technical terms such as drug names or active ingredient codes. The top-10 candidate chunks are retrieved.

LLM-as-Judge reranking. Claude 3.5 Sonnet evaluates the relevance of each retrieved chunk with respect to the original question, selecting the top-3 chunks for context construction. This reranking step reduces noise in the prompt and improves response quality.

CRAG relevance evaluation. The relevance of the selected context is evaluated against a minimum threshold. If relevance is insufficient, the query is reformulated and a second retrieval cycle is triggered. If context quality remains below threshold after the second attempt, the system activates a low-confidence response mechanism: it refuses to generate a response and informs the user that the question cannot be answered based on available documentation, recommending consultation of a qualified healthcare professional.

Response generation. When context is deemed sufficient, an enriched prompt combining the query, retrieved chunks, and source metadata is submitted to Claude 3.5 Sonnet. The model generates a response grounded exclusively in the provided context, including explicit citation of source document, section, and page.

Output guardrail. The generated response is verified for faithfulness to the retrieved context by a second LLM-as-Judge evaluation. Responses below the faithfulness threshold are flagged with an explicit warning. Additionally, all responses include a mandatory disclaimer stating that the information does not substitute the advice of a qualified healthcare professional. The validated response is returned via the API and the full interaction is logged.

E. Technology Stack and Implementation Status

Table II summarises the technology stack and the rationale for each selection.

Gemini Embedding 2 was preferred over OpenAI text-embedding-3-small due to its native multilingual support and direct PDF document encoding, critical for the Portuguese pharmaceutical corpus. Qdrant’s native hybrid retrieval and local Docker deployment satisfy both the hybrid

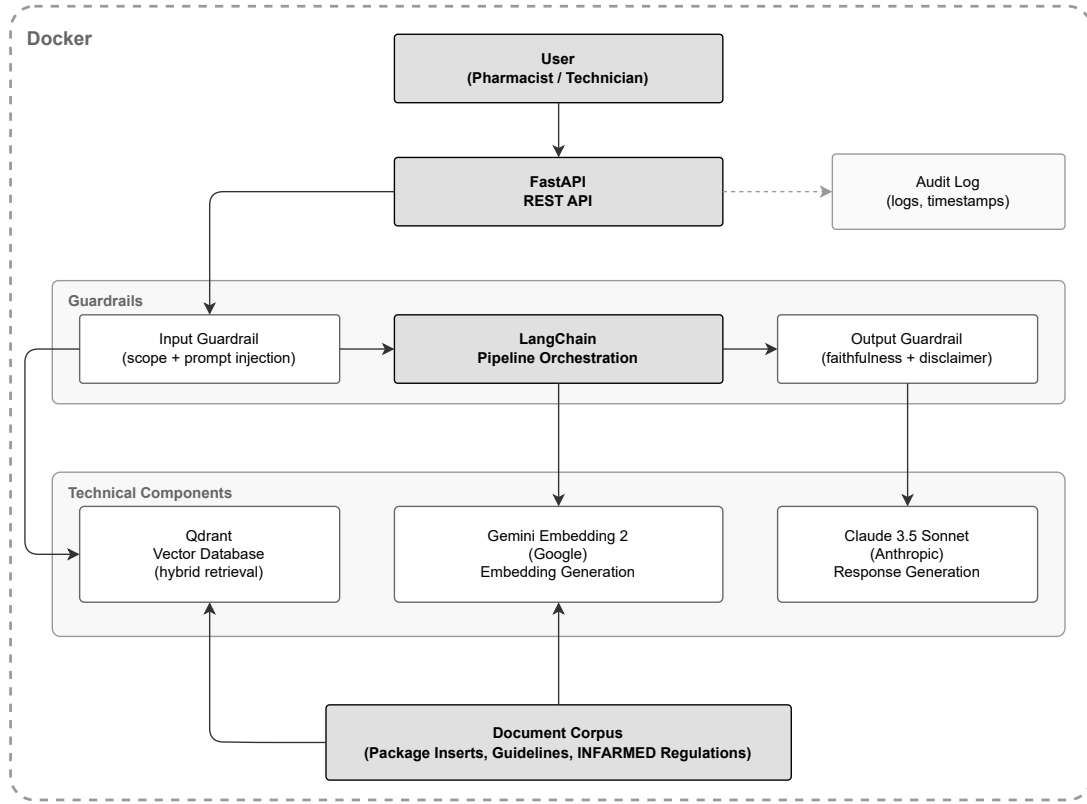


Fig. 1. System architecture overview. The system is organised in three functional layers — input/control, orchestration, and knowledge base — with audit logging running transversally across all layers.

retrieval requirement and GDPR data locality constraints. Claude 3.5 Sonnet was selected for its reported lower hallucination rate in document-grounded QA tasks and its Constitutional AI design, aligned with the system’s high-risk classification.

The ingestion pipeline has been implemented and validated on a preliminary corpus, producing a stable indexing process with consistent metadata preservation across PDF document types. The query pipeline implementation is in progress at the time of writing, with the evaluation phase scheduled upon its completion.

V. EVALUATION

A. Evaluation Methodology

System performance is assessed using the RAGAS framework [8] across the four metrics described in Section II: faithfulness, answer relevancy, context precision, and context recall. The evaluation set consists of N question-answer pairs representative of real pharmaceutical usage scenarios, including queries about dosage, contraindications, drug interactions, and regulatory information, constructed from official INFARMED documentation.

Three experimental conditions are evaluated: (i) the full proposed system with hybrid retrieval, LLM-as-Judge reranking,

and CRAG correction; (ii) an ablation without reranking (top-10 chunks passed directly to generation); and (iii) an ablation using semantic-only retrieval without keyword search. This ablation design allows direct quantification of the contribution of each architectural component.

B. Results

Results to be completed upon finalisation of the evaluation phase.

C. Discussion

Discussion to be completed with quantitative results.

VI. CONCLUSION AND FUTURE WORK

This paper presented a RAG-based system for pharmaceutical decision support in the Portuguese regulatory context, addressing three gaps identified in the systematic literature review: the scarcity of biomedical RAG systems adapted to non-English contexts and national proprietary documentation; the absence of EU AI Act compliance in existing systems; and the limited use of corrective retrieval mechanisms in pharmaceutical applications.

The system architecture combines Advanced RAG, Modular RAG, and CRAG, implemented with Claude 3.5 Sonnet, Gemini Embedding 2, Qdrant, LangChain, and FastAPI, and

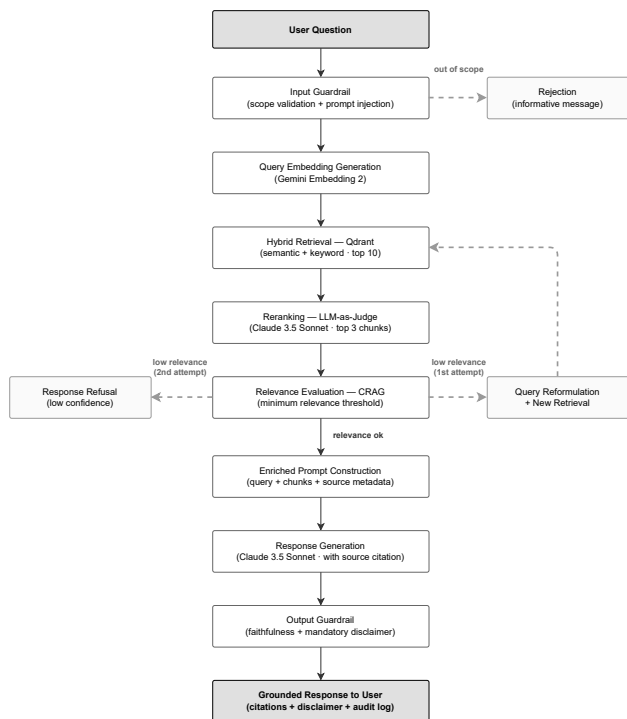


Fig. 2. Query pipeline. Solid arrows indicate the main flow; dashed arrows indicate alternative branches triggered by guardrails or by the CRAG relevance evaluation.

deployed locally via Docker in conformance with GDPR data locality requirements. Safety guardrails at both input and output stages, mandatory source citation, and a comprehensive audit log ensure alignment with the EU AI Act requirements for high-risk AI systems.

Several directions for future work are identified. First, extending the knowledge base to cover the full INFARMED medication database, beyond the initial corpus used for evaluation. Second, investigating domain-adapted embedding models trained specifically on Portuguese pharmaceutical text, to improve retrieval of highly specialised terminology. Third, exploring GraphRAG approaches to model the relational structure of pharmacological knowledge — drug interactions, therapeutic groups, and contraindications — which is inherently graph-structured. Finally, integrating multimodal processing to handle chemical structure images and pharmacokinetic graphs present in technical data sheets, currently discarded by the ingestion pipeline.

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